Comparative Analysis of Lion and AdamW Optimizers for Cross-Encoder Reranking with MiniLM, GTE, and ModernBERT

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Introduction

Research Overview & Motivation

- **Information Retrieval Challenge:** Modern IR systems require precise document ranking from vast collections
- **Cross-Encoder[2] Advantage:** State-of-the-art reranking through deep query-document interactions
- **Optimization Gap:** Limited research on Lion optimizer[4] for IR tasks despite success in other domains
- **Practical Impact:** Optimizer choice significantly affects training cost and model performance

Research Problem:

How do Lion[4] and AdamW[3] optimizers compare across different cross-encoder architectures for information retrieval tasks?

Research Objectives

1. **Systematic Evaluation:** Compare Lion[4] and AdamW[3] across three transformer architectures.
2. **Comprehensive Benchmarking:** Evaluate performance using standard IR metrics on TREC DL 2019[8] and MS MARCO[9].
3. **Training Dynamics Analysis:** Investigate convergence patterns and stability.
4. **Practical Guidelines:** Provide evidence-based recommendations for optimizer selection.

Expected Contributions

- **Empirical evidence comparing optimizers across multiple architectures**
- **Optimization guidelines for cross-encoder training in IR**
- **Analysis of training efficiency and resource utilization**

Literature Review

BERT & Cross-Encoder Architecture

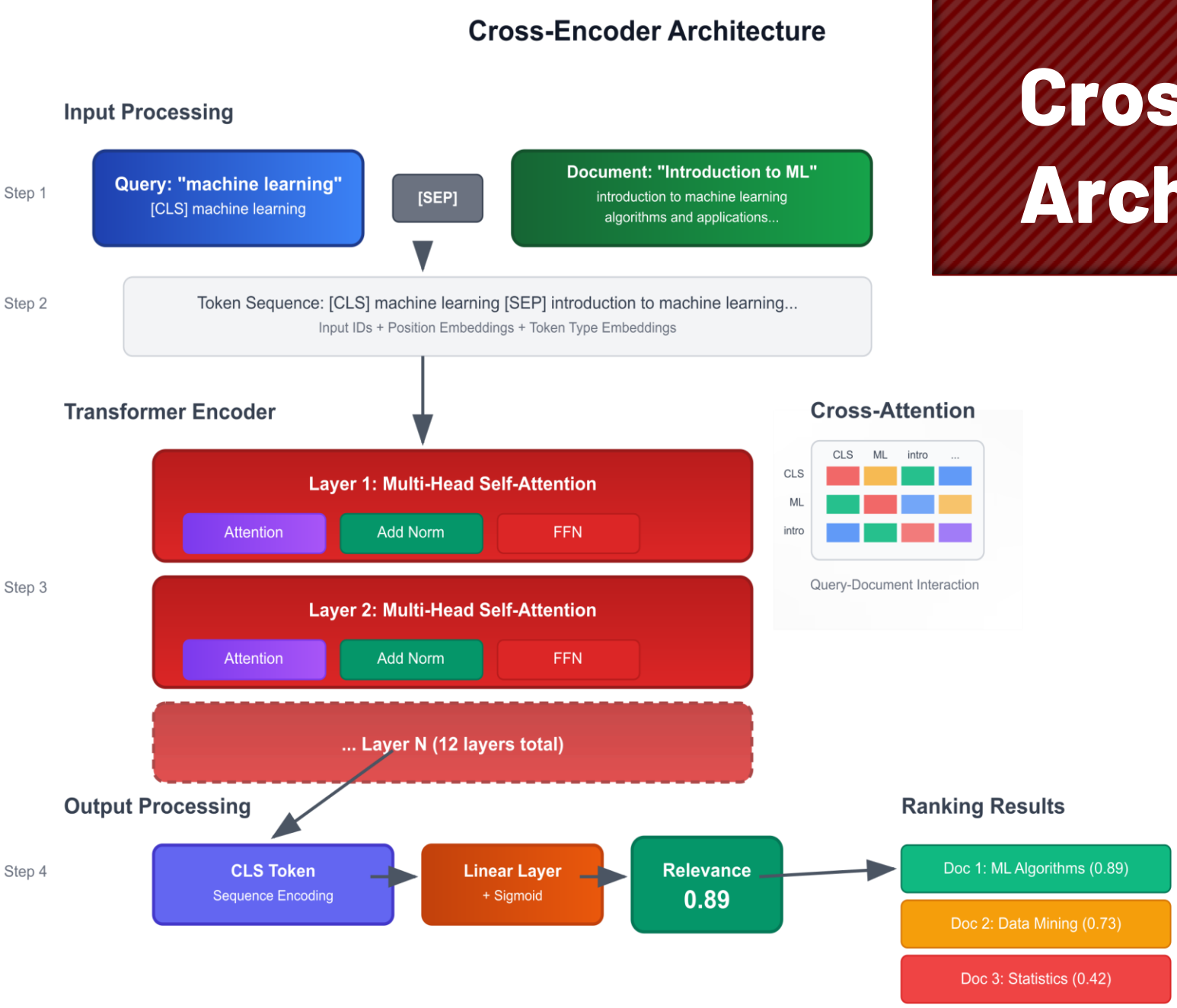
BERT-based Cross-Encoders[2] for Information Retrieval

Key Innovation:
- Joint encoding: [CLS] Query [SEP] Document [SEP]
- Fine-tuning for relevance prediction
- Superior to bi-encoders for reranking

Modern Variants:
- MiniLM[5]: Distilled BERT for efficiency
- GTE[7]: Multilingual, extended context (8192 tokens)
- ModernBERT[6]: Advanced features (RoPE[12], Flash Attention[13])

Standard approach for document reranking tasks

Cross-Encoder Architecture



Key Advantages

- ❑ Deep Interaction
- ❑ Token-level Attention
- ❑ Superior Accuracy

Image Source: Claude Sonnet 4 (Anthropic, 2025)

AdamW: The Standard Optimizer [3]

Evolution: Adam → AdamW

Key Improvement:
• Decoupled weight decay from gradient updates

Why AdamW Dominates:
✓ Better generalization than Adam
✓ Stable training for transformers
✓ De facto standard for BERT[1] family models

Current Status: Default choice without alternatives explored

AdamW Algorithm

AdamW Optimizer

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \tag{1}$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \tag{2}$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t} \tag{3}$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t} \tag{4}$$

$$\theta_t = \theta_{t-1} - \eta \left(\frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} + \lambda \theta_{t-1} \right) \tag{5}$$

where:

g_t: gradient at time step t

m_t: exponential moving average of gradients (momentum)

v_t: exponential moving average of squared gradients

η: learning rate

λ: weight decay coefficient

β_1, β_2: exponential decay rates for moment estimates

Lion: A Novel Alternative[4]

Revolutionary Approach:
- Sign-based momentum
- Simplified computation vs Adam family
- Reduced memory requirements

Proven Results:
✓ Competitive performance in computer vision
✓ Lower computational overhead
✓ Faster convergence in some tasks

Research Gap:
❌ Unexplored in NLP/Information Retrieval
❌ No systematic comparison with AdamW for cross-encoders

Lion Algorithm

Lion Optimizer

$$c_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \tag{1}$$

$$\theta_t = \theta_{t-1} - \eta \left(\text{sign}(c_t) + \lambda \theta_{t-1} \right) \tag{2}$$

$$m_t = \beta_2 m_{t-1} + (1 - \beta_2) g_t \tag{3}$$

where:

c_t: interpolated momentum for update direction

$\text{sign}(\cdot)$: element-wise sign function

Other parameters follow similar definitions as AdamW

Methodology

Experimental Setup

Dataset Configuration

Training: MS MARCO[9] Passage Ranking (~500K query-passage pairs)

Evaluation: TREC DL[8] 2019 (43 queries with comprehensive relevance judgments)

Metrics: NDCG@10, MAP, MRR@10, Recall@10, R-Precision, P@10

Infrastructure

Platform: Modal cloud computing[10]

Hardware: 3 × NVIDIA L40S GPUs

Monitoring: Weights & Biases[11]

Training: 3 epochs per configuration

Experimental Design

Total Configurations: 6 (3 models × 2 optimizers)

Controlled Variables: Batch size, training steps, evaluation frequency

Variable Parameters: Learning rates (2e-5 standard, 2e-6 for ModernBERT+Lion)

Model Architectures Tested

Model	Parameters	Context Length	Key Features
MiniLM-L12-H384	~33M	512 tokens	Efficient baseline, knowledge distillation
GTE-multilingual-base	~305M	8192 tokens	Extended context, multilingual support
ModernBERT-base	~149M	8192 tokens	RoPE, Flash Attention, GeGLU

MiniLM: Efficiency-focused for resource-constrained scenarios

GTE: Extended context for comprehensive document analysis

ModernBERT: State-of-the-art with advanced architectural innovations

⚙ Hyperparameter Settings

Parameter	AdamW	Lion
Learning Rate	2e-5 (standard)	2e-5 (2e-6 for ModernBERT)
Weight Decay	0.01	0.01
Beta1	0.9	0.9
Beta2	0.999	0.99
Scheduler	Linear	Cosine Annealing[15] (ModernBERT)

Results & Discussion

Main Results

Model	Optimizer	NDCG@10	MAP	MRR@10	Recall@10	P@10
MiniLM	AdamW	**0.7127**	**0.4908**	0.5826	0.1715	0.8093
	Lion	0.7031	0.4858	**0.5988**	0.1724	0.8116
GTE	AdamW	**0.7224**	**0.5005**	0.5942	**0.1733**	0.8163
	Lion	0.6909	0.4921	0.5957	0.1721	0.8140
ModernBERT	AdamW	0.7105	0.5066	0.5916	0.1678	0.8163
	Lion	🏆**0.7225**	🏆**0.5121**	🏆**0.5988**	0.1732	🏆**0.8256**

🏆 **Overall Winner: ModernBERT + Lion achieved highest performance in 4/5 metrics**

GPU Usage Statistics

Model	Optimizer	Mean Usage (%)	Peak Usage (%)	Efficiency gain (%)
MiniLM	AdamW	33.09	35.25	+2.67 %
	Lion	32.21	34.64	
GTE	AdamW	73.04	78.08	+10.33 %
	Lion	65.50	69.69	
ModernBERT	AdamW	77.04	81.40	+3.49%
	Lion	74.35	79.45	

Lion optimizer demonstrates superior GPU utilization efficiency across all models

Key Takeaways & Decision Points

🎯 Major Findings Summary

- 🏆 **Universal Efficiency:** Lion optimizer reduces GPU usage in 100% of configurations
- ⚖️ **Performance Trade-offs:** Neither optimizer dominates all metrics
- 🔧 **Configuration Critical:** Learning rate and scheduling significantly impact results
- 🚀 **ModernBERT Excellence:** Best overall performance with Lion optimizer
- 💰 **Cost-Effectiveness:** 5.50% average efficiency improvement across models

🧠 Decision Matrix for Practitioners

Priority	Model Choice	Optimizer	Expected Outcome
Maximum Performance	ModernBERT	Lion	Best metrics + efficiency
Cost Optimization	GTE	Lion	10.33% resource savings
Balanced Approach	MiniLM	Lion	Good performance + efficiency
Risk Minimization	Any	AdamW	Established, reliable results

Future Research Directions

Immediate Extensions

- Comprehensive hyperparameter studies
- Extended training analysis
- Alternative schedulers investigation
- Cross-lingual evaluation

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